



Society of Petroleum Engineers

SPE-225248-MS

Integrated Process and Methods in Identifying ESP Trip Root Cause

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This paper was prepared for presentation at the SPE Gulf Coast Section - Electric Submersible Pumps Symposium held in Galveston, Texas, USA, 25–29 August 2025.

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Abstract

This paper outlines an integrated process and methods in troubleshooting Electrical Submersible Pump (ESP) downtime by identifying the trips root cause through a systematic approach whereas more than 15 surface and subsurface parameters are thoroughly analysed. Moreover, the paper discussed how the behaviour and markers of these combined parameters can result in each ESP trip scenario. The best applicable remedy actions are laid out for each separate case taking into considerations maintaining ESP system mechanical and electrical integrity.

More than 15 surface and subsurface related parameters are thoroughly analysed and the behaviour of the combined effects of each parameter on each other is utilized to identify the multiple trip markers. Through trip markers analysis, more than 10 general trip causes were classified with a clear process and workflow to distinguish each case during the troubleshooting process. Moreover, specific causes in each general case are also discussed in this paper to further distinguish and identify the exact trip cause. The paper also outlines the recommended remedy action to be taken to return the ESP back online with minimized downtime.

ESP trip is defined as any unwanted shutdown of ESP system through the activation of ESP drive protection setpoint due to internal or external fault or situation. Any component in the production system from reservoir up-to processing facilities can impact ESP systems negatively. Similarly, any component in the electrical system from motor up-to electrical substation can impact it with the same intensity as well. For these reasons, multiple factors can lead to an ESP trip leading to a time-consuming process to identify the root cause and returning the ESP online. The results discussed in this paper shows that the exact root cause can be promptly identified through the mentioned methodology with high accuracy leading to successful restarting of ESP system in optimized operating point and conditions.

The results and conclusions discussed in this paper can be utilized to successfully troubleshoot any tripped ESP by accurately identifying the exact root cause through the discussed markers. The recommended remedy actions explained in this paper ensures the ESP is back online with minimal downtime with an optimized performance that maintains its electrical and mechanical integrity.

Keywords: Artificial Lift, ESP, gas-lock, ESP run-life, downtime, ESP trips, ESP failure, data analysis

Introduction

ESP Failure

ESP systems are composed of various components with mechanical and electrical aspects. These components contain multiple elements each with different objective and purpose. As such, ESP systems can fail due to multiple reasons and by many casual failure factors. Identifying ESP failure root causes is done through a comprehensive dismantle, inspection, and failure analysis (DIFA) process (Pessoa, et al. 2024).

Electrical Failure. Electrical failures occur when the motor fails, not being able to fully or partially spin due to compromised electrical integrity in any part of the ESP electrical system. Most commonly due to phase-to-ground faults or phase-to-phase faults. In such conditions the power supplied will not met the threshold necessary for the motor to provide the required load; and motor voltage and current will be unbalanced across the three-phases and/or vary in multiple folds above or below the motor rated specifications.

Mechanical Failure. Mechanical failures occur when the mechanical energy is not transferred fully or partially from the motor to the pump or any part thereof. It can be sub-classified further into below main classifications.

Stuck Shaft

This type of mechanical failure occurs when the motor fails to rotate due to the shaft or any other mechanical component getting stuck. This results in motor drawing high amperage until the shaft is released, breaks, or the system fails electrically due to the high electrical stress.

Broken Shaft

This type of mechanical failure occurs when the motor rotates but the mechanical energy is not transmitted to the pump due to broken mechanical components or mechanical disengagement between them. The motor continues to rotate but operates in underload conditions as no load is exhibited in the motor (inability to lift fluids).

Underperformance

This type of mechanical failure occurs when the flow at surface is not as designed based on pump performance curve due to mechanical underperformance in the ESP system. Most commonly occurs in aged systems or those in gassy or sandy environments as the stages get damaged over time thus losing their efficiency (Boudi, et al. 2024).

Pump failure

This occurs when the pump and all associated parts (stages; diffusers and/or impellers) get damaged and unable to lift the required fluids. This mode of failure can be expressed in different scenarios of completely wear of pump stages (fully or partially) as in the case of diffusers spinning or impellers non-spinning. Such events might occur due to: manufacturing defects or foreign/incompatible fluid type productions.

ESP Trip

ESP trip is defined as any unwanted shutdown of ESP system through the activation of ESP drive protection setpoint due to internal or external fault or situation. Thus, any shutdown other than manual ESP shutdown - locally or remotely - is considered an ESP trip. ESP trips vary in complexity based on the root cause of the trip from basic trips that ESP can be restarted after with no changes in parameters to very complex trips that require significant enhancements and optimization in ESP operating conditions. To recognize which part of the spectrum each trip falls under and the required steps to overcome, the most critical item is to identify the exact root cause of the trip. This not only ensures production continuity and reduce deferred revenue due to offline ESP system but also ensures the ESP integrity is maintained by reducing the repetitive trips and avoiding operating the ESP system in unfavorable conditions.

Protection Setpoints

Protection Setpoints are user-defined settings in VSD to allow for automatic ESP shutdown without human intervention under specific scenarios. They are commonly called "Trip Setpoints" as they lead to an ESP trip. The term "Trip" means any ESP shutdown due to unforeseen conditions. Basically, all ESP shutdowns other than Planned Shutdown -- manual local or remote stop. These setpoints ensure healthy ESP integrity as operating the ESP system in such conditions leads to ESP failure. There are many Protection Setpoints but they all primarily maintain ESP integrity by targeting the following two conditions.

Motor Overheat. The primary mean of cooling the motor during its operation is the heat transfer between the motor and the produced fluid. Lack of flow across the motor leads to significant motor -and other ESP components- overheat and mechanical wear due to operating outside the pump recommended operation range (RoR).

Electrical Overload. The motor windings are rated for specific amperage based on the motor type and model. Exceeding the nameplate amperage compromises the electrical integrity of the system due to significant overheat of the electrical components.

Motor Load

Motor Load is defined as the percentage of operating motor horsepower in reference to its nameplate horsepower. However, since amps values are readily available and continuously measured by the drive and since the relationship between HP load $\left(\frac{\text{Operating HP}}{\text{ameplate HP}} \times 100\right)$ and amps load $\left(\frac{\text{Operating Amps}}{\text{Nameplate Amps}} \times 100\right)$ is relatively linear and nearly 1:1 with HP load at higher loads then the motor load definition is used as the latter not the former.

During ESP design and operation, the common practice is to have the minimum motor load to be 50%. This is because the efficiency and power factor of the system drops significantly below 50%. The absolute minimum motor load is the idle motor load which is around 30%-40% motor load (Williams 2019).

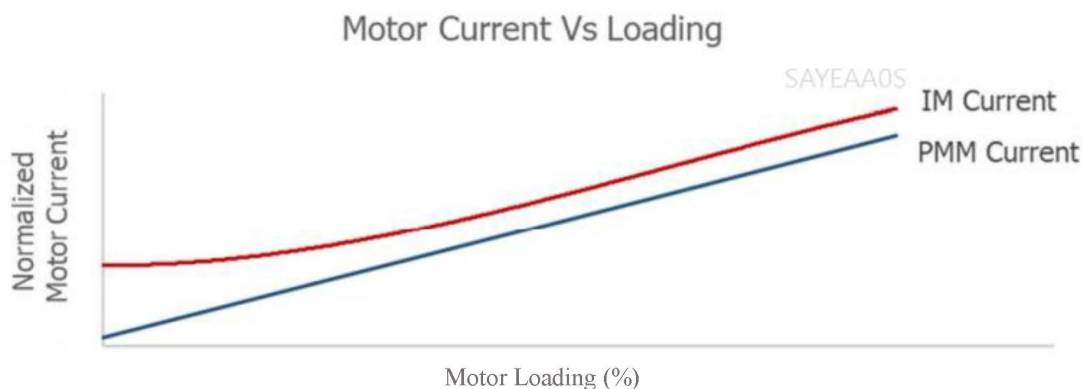


Figure 1—Example of general relationship between electrical motor current versus loading (Gad, et al. 2025).

Underload

The motor converts the electrical energy to mechanical energy. The mechanical energy is transferred from the shaft onto the pump shaft and then is imparted on the produced fluid. Hence, the amount of required electrical energy by the motor is directly proportional and dependent of the generated hydraulic energy. If there is no flow generated by the pump the motor load would be much less and thus is "underloaded". The underload setpoint is commonly set at 80%-90% of running amps (Camilleri, et al. 2018) or amps value during deadhead condition. Most common causes for Underloading the motor: Blockage in flow path above pump discharge (scale build-up, downhole plug/BPV, closed SSSV/SSV, closed valve in wellhead or flowline), gas lock, Broken Shaft, Damaged Stages, Wrong Rotation, etc.

Pump Discharge Pressure

Pump discharge pressure is inversely proportional to flowrate and directly proportional to produced fluid specific gravity and frequency. Thus, an increase in discharge pressure is a common indication of no-flow across the motor hence leading to motor overheating. The discharge pressure setpoint is commonly set at 150 psi - 300 psi above running discharge pressure. Most common causes for High Discharge Pressure: Blockage in flow path above pump discharge (scale build-up, downhole plug/BPV, closed SSSV/SSV, closed valve in wellhead or flowline), changes in produced fluid composition, etc.

Motor Temperature

Heat transfer with fluid flowing across the motor is the primary cooling mechanism for the ESP motor and with no flow (or less flow than anticipated) the motor will significantly overheat and burn. The motor temperature setpoint depends on motor and dielectric oil types. Most common causes for high motor temperature: Blockage in flow path above pump discharge (scale build-up, downhole plug/BPV, closed SSSV/SSV, closed valve in wellhead or flowline), changes in produced fluid composition, gas lock, running with higher motor current values (e.g. higher feed amperage) etc.

Overload

Motor load is directly proportional to produced flowrate -- and generated head. While higher flowrate leads to better cooling of the motor there will be a tipping point whereas the increase in frequency and increase in running amps will generate more heat leading to motor overheating. Running frequency, fluid specific gravity, and system electrical integrity all can lead to overload as well. The overload setpoint is commonly set at 110%-115% of nameplate amps ([Awaid, et al. 2014](#)) and can be set to maximum of 120% for troubleshooting purposes. Most common causes for Overload: High running frequency, wrong sized motor, stuck shaft, changes in produced fluid composition, compromised system electrical integrity.

Discussion and Results

This section discusses the common ESP trip root cases and the procedures recommended to be employed while troubleshooting ESP systems and prior to any ESP failure declaration.

Common Trips

Case #1.

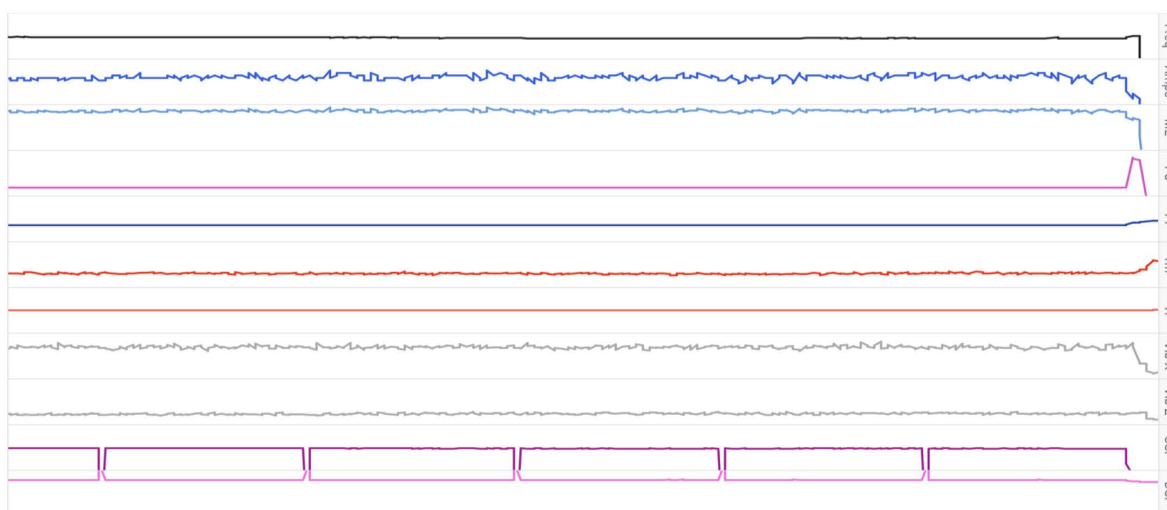


Figure 2—ESP trip due to blockage above ESP.

Diagnosis

Blockage Above Pump.

Markers

Blockage Above Pump: increase in discharge pressure.

Reduction in Flow to Surface: decrease in amperage and wellhead pressure; increase in intake pressure and motor temperature.

Remedy

Check the downhole completion schematics and surface layout. Ensure there are no downhole plugs are set or valves can be closed. Similarly, check the valves on the surface flowline.

Table 1—Blockage above ESP parameters trend.

Parameter	Trend
Discharge Pressure	Increase
Amps	Decrease
Motor Temperature	Increase
Wellhead Pressure	Decrease

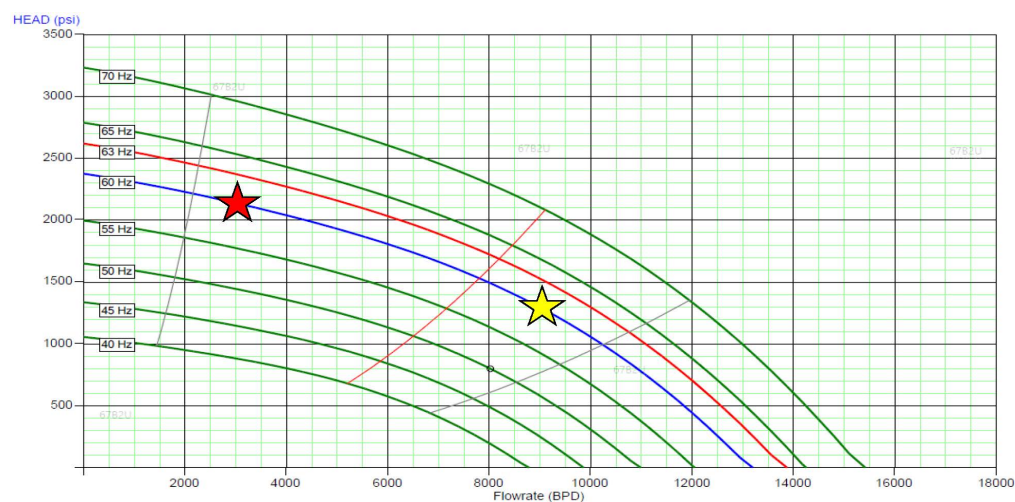


Figure 3—ESP operating point before (yellow) and after (red) the blockage above pump incident.

Case #2.

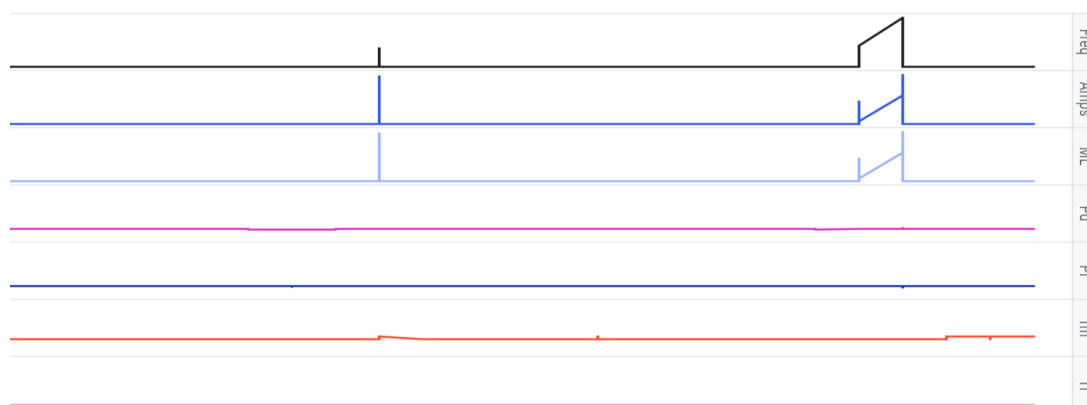


Figure 4—ESP trip due to stuck shaft.

Diagnosis

Stuck Shaft

Markers

Shaft Difficult to Rotate: high amps exceeding motor nameplate at low frequency, increase of motor and fluid intake ambient temperatures

Remedy

No-Load test to confirm drive healthiness.

Integrity test to confirm downhole system healthiness.

Increase start-up torque through VSD hard-start or rock-start methods.

Chemical injection to eliminate scale/solids accumulation.

Table 2—Stuck Shaft parameters trend.

Parameter	Trend
Amps	Increase
Frequency	Remains Low

Case #3.

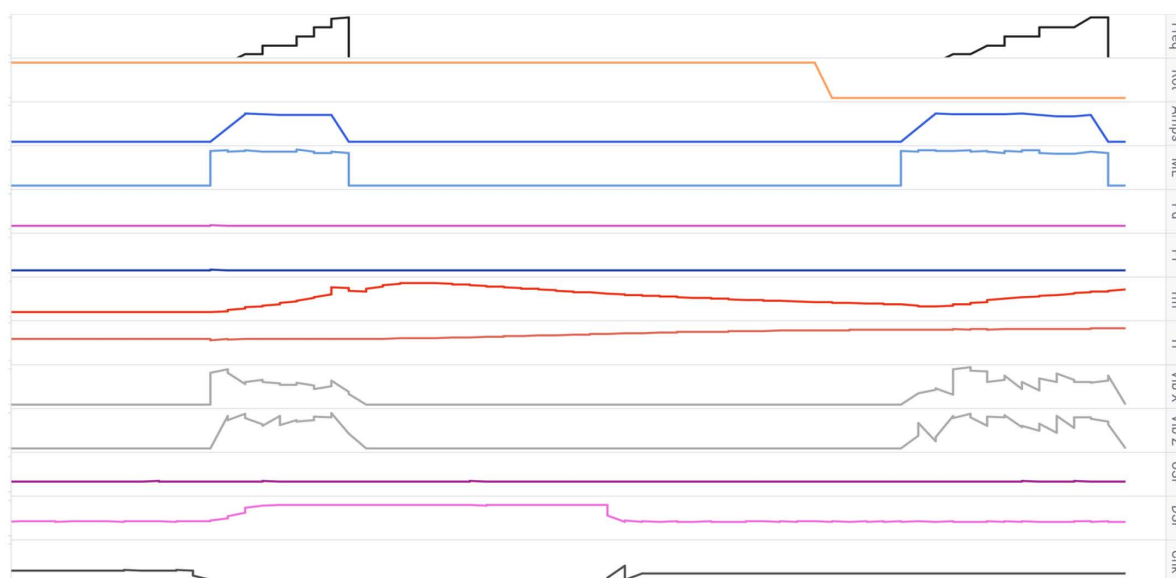


Figure 5—ESP trip due to broken shaft.

Diagnosis

Broken shaft below the pump.

Markers

No Load on Motor: Very low motor amperage; motor load near idle conditions; frequency changes don't affect the amperage; motor draws same amperage at both rotations.

No flow: increase in motor temperature; no change in intake temperature, equalized intake pressure and discharge pressure, and static wellhead pressure.

Mechanical Issue: high fluctuations in vibration.

Remedy

Follow Failure Declaration Workflow and arrange for ESP system replacement.

Table 3—Broken Shaft parameters trend.

Parameter	Trend
Amps	Decrease
Discharge Pressure	Decrease
Wellhead Pressure	Decrease
Motor Temperature	Increase

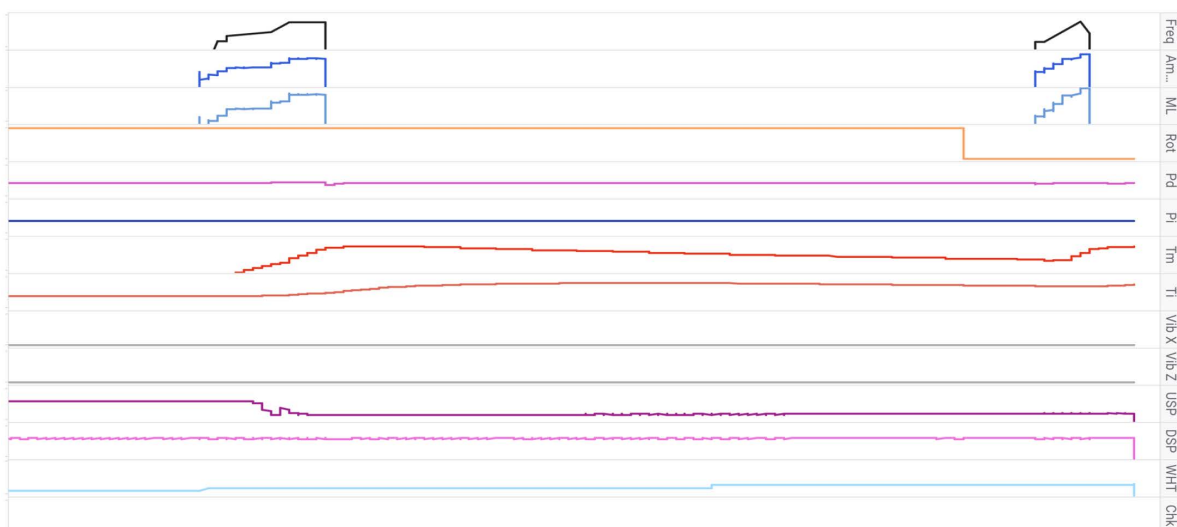
Case #4.

Figure 6—ESP trip due to full downhole circulation.

Diagnosis

Full Downhole Circulation.

Markers

No Flow to Surface: Low WHP; Pd equal to Pi; no surface flowrate; low WHT.

Downhole Circulation: Low differential pressure; high Tm; significant increase in Tm after shutdown.

Healthy Shaft: High motor amps and load; amps react to changes in frequency.

Remedy

Rigless Job: Replace Y-Tool through WL.

Rig Job: Replace Downhole tubing and completion.

Table 4—Full Downhole Circulation parameters trend.

Parameter	Trend
Discharge Pressure	Decrease
Wellhead Pressure	Decrease
Motor Temperature	Increase

Case #5.

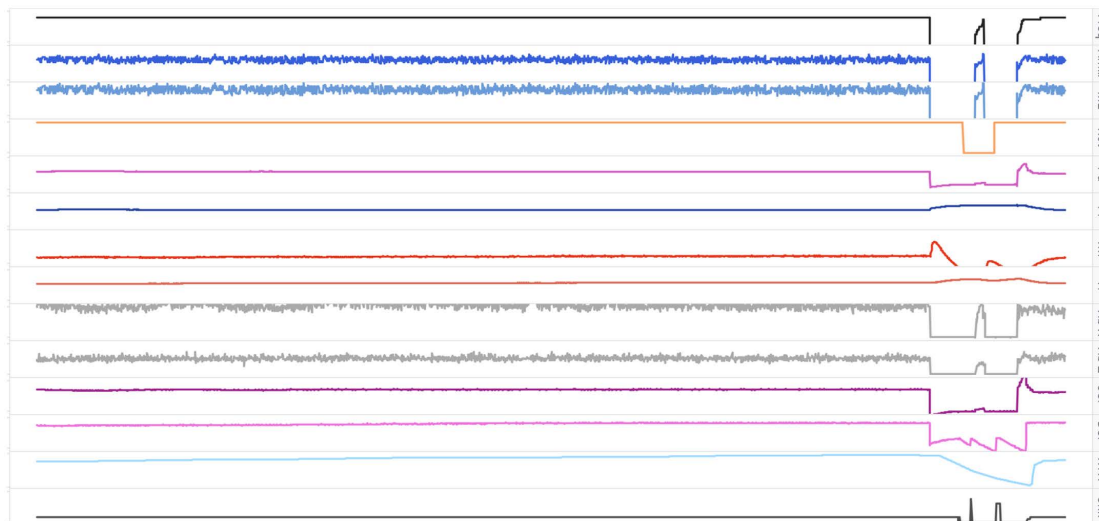


Figure 7—ESP trip due to partial downhole circulation.

Diagnosis

Partial Downhole Circulation.

Markers

Low Flow to Surface: Low WHP; slight Pd increase; low surface flowrate; low WHT.

Downhole Circulation: Low differential pressure; high Tm; significant increase in Tm after shutdown.

Healthy Shaft: High motor amps and load; amps react to changes in frequency.

Remedy

Rigless Job: Replace Y-Tool through WL or CT.

Rig Job: Replace Downhole tubing and completion.

Applying Fluid Hammering technique. In this procedure the wing valve will be closed suddenly while the ESP is operating at high frequency thus creating a back-pressure shock transferring all the fluid momentum back into the Y-Tool plug at setting it in-place

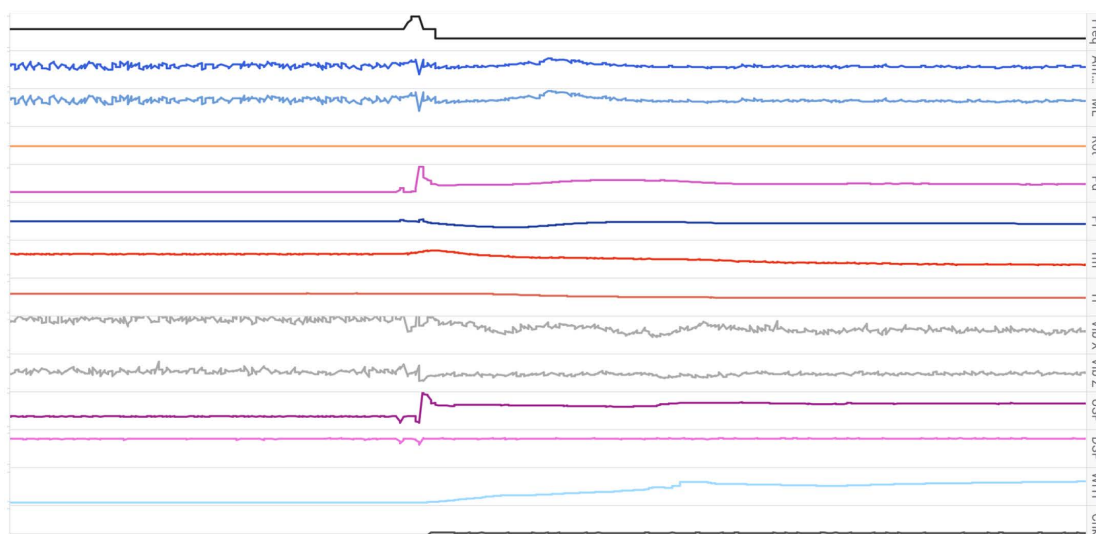


Figure 8—Example of successful Fluid Hammering application.

Table 5—Example of running parameters before and after applying fluid hammering.

Parameter	Before	After
Freq	53	56
Amps	42	41
Pi	2408	2507
Pd	3200	2865
Tm	282	284
FWHP	970	670
Choke	12%	10%

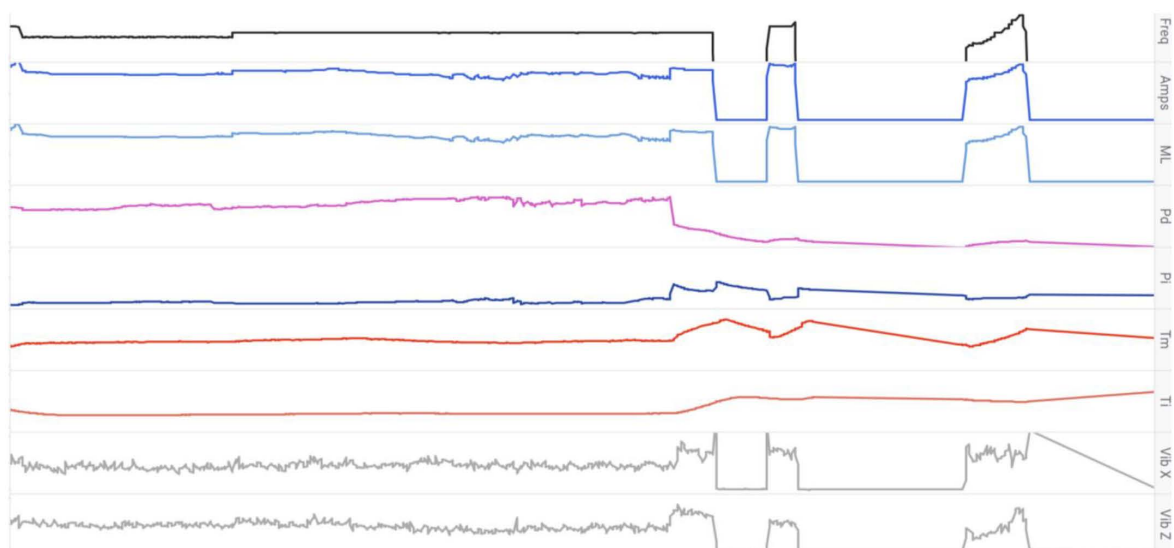
Case #6.

Figure 9—ESP trip due to circulation above ESP and below packer.

Diagnosis

Downhole Circulation: Ruptured Tubing Below Packer.

Markers

No Flow at Surface: low WHP; increase in Tm; significant increase in Tm after shutdown.

High Downhole Rate: low differential pressure; high motor amps and motor load.

Healthy Shaft: high motor amps and motor load; amps react to changes in frequency.

Remedy

Mechanical Workover to replace tubing and completion.

Table 6—circulation above ESP and below packer parameters trend.

Parameter	Trend
Discharge Pressure	Decrease
Wellhead Pressure	Decrease
Motor Temperature	Increase

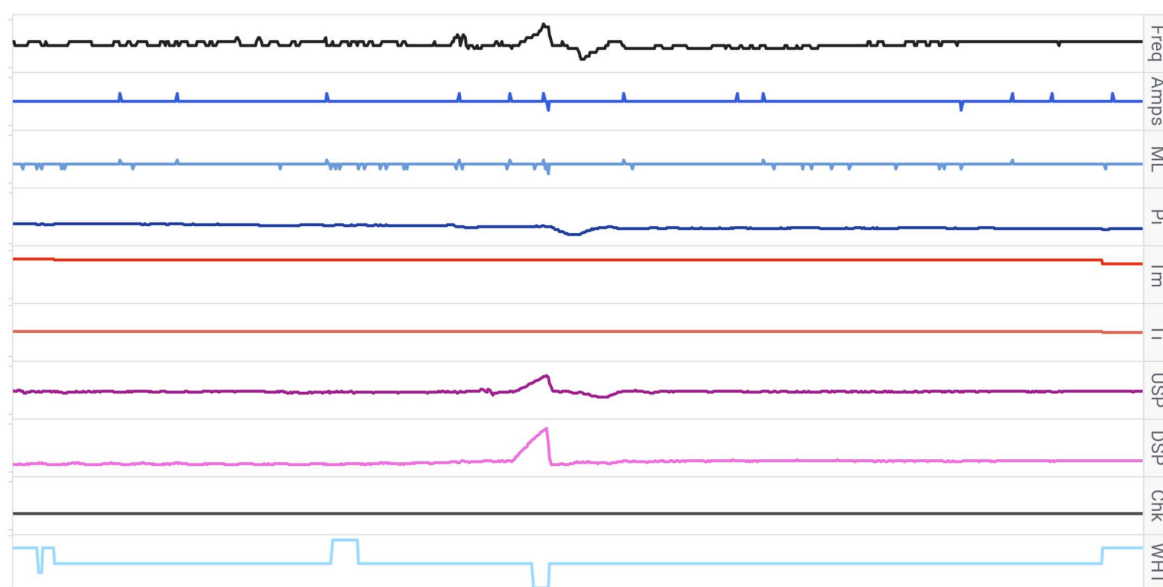
Case #7.

Figure 10—ESP trip due to blockage in flow path.

Diagnosis

Blockage Downstream of Choke.

Markers

Blockage Downstream of Choke: Increase in downstream pressure.

Low Flow: Decrease in motor amps and motor load; decrease in wellhead temperature; increase in motor temperature.

Blockage Above ESP: Increase in discharge pressure.

Remedy

Check on surface flowline line-up.

Increase differential pressure across choke to prevent downstream pressure from controlling the rate.

Adjust trip setpoints accordingly.

Table 7—blockage in flow path parameters trend.

Parameter	Trend
Discharge Pressure	Increase
Wellhead Pressure	Increase
Motor Temperature	Increase
Amps	Decrease

Case #8.

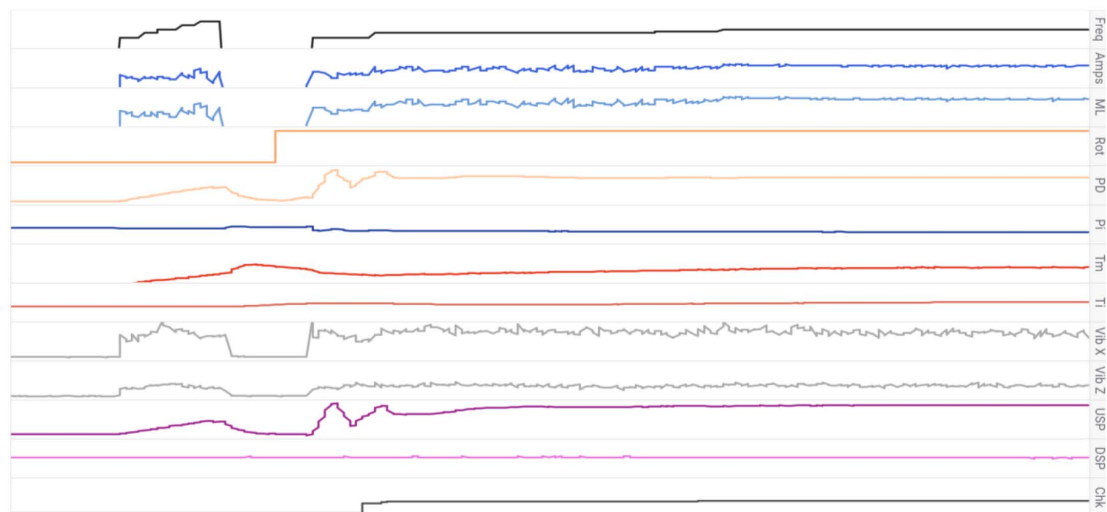


Figure 11—ESP trip due to wrong rotation.

Diagnosis

Wrong Rotation.

Markers

Significantly Lower Flowrate: Low Pd; low WHP; low WHT; low motor amps and load; increase in Tm.

Remedy

Conduct deadhead test at both rotations.

Table 8—wrong rotation parameters trend.

Parameter	Trend
Discharge Pressure	Decrease
Wellhead Pressure	Decrease
Motor Temperature	Increase
Amps	Decrease

Case #9.

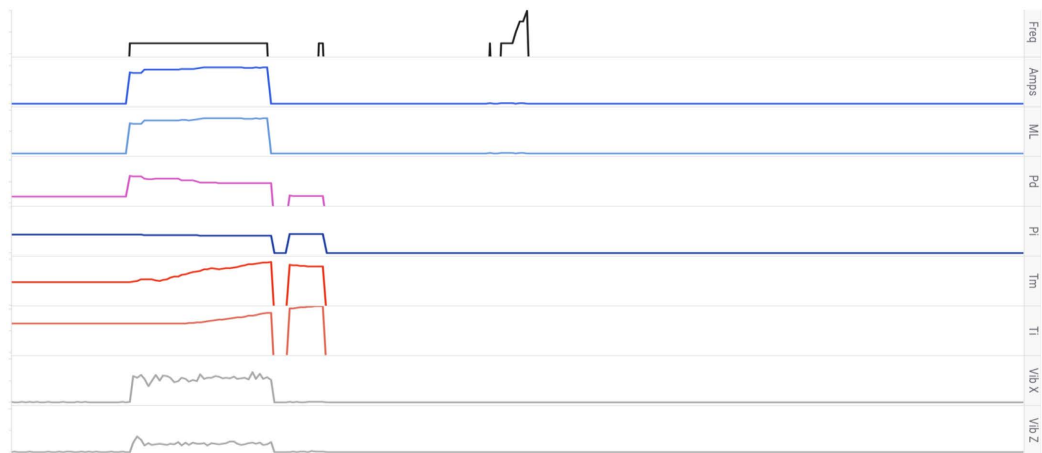


Figure 12—ESP trip due to no load.

Diagnosis

No Load: Penetrator Cable Disengagement.

Markers

No Load: Zero amps.

Remedy

Immediately shutdown and check all cable connections.

Check downhole system integrity.

If all still good, cross-check the amps through a multi-meter.

Table 9—no-load parameters trend.

Parameter	Trend
Amps	Zero

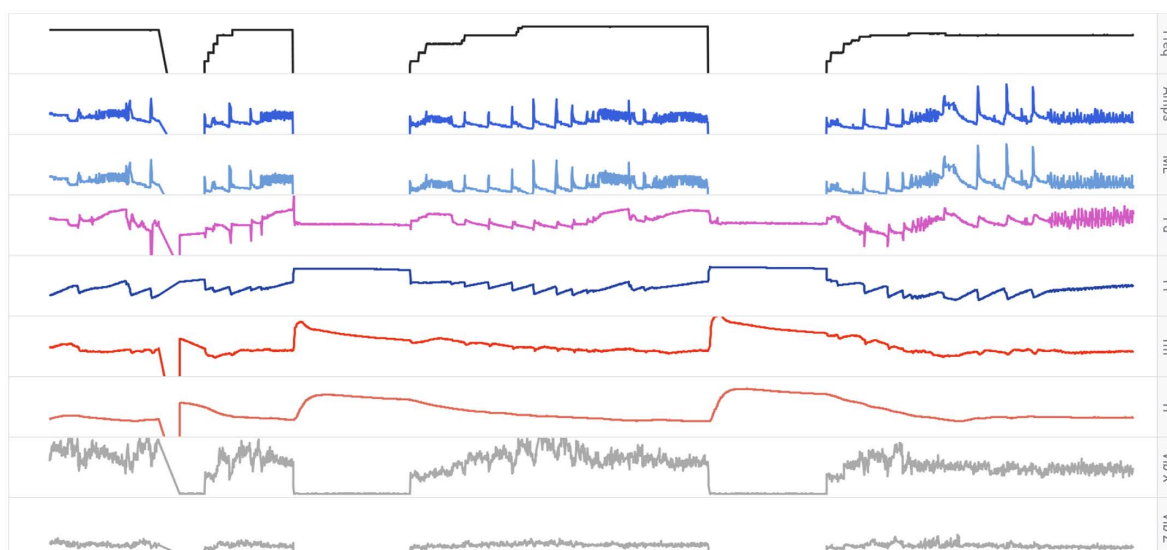
Case #10.

Figure 13—ESP trip due to pump off conditions.

Diagnosis

Pump-Off: Oversized Pump/Low PI.

Markers

Significant decrease in Pi, diminishing flow ate surface, downhole instabilities, high Tm.

Remedy

Operate pump at left hand side.

Mechanical Workover to replace ESP based on the actual parameters.

Table 10—pump-off parameters trend.

Parameter	Trend
Intake Pressure	Decrease
Wellhead Pressure	Decrease
Motor Temperature	Increase

Case #11.

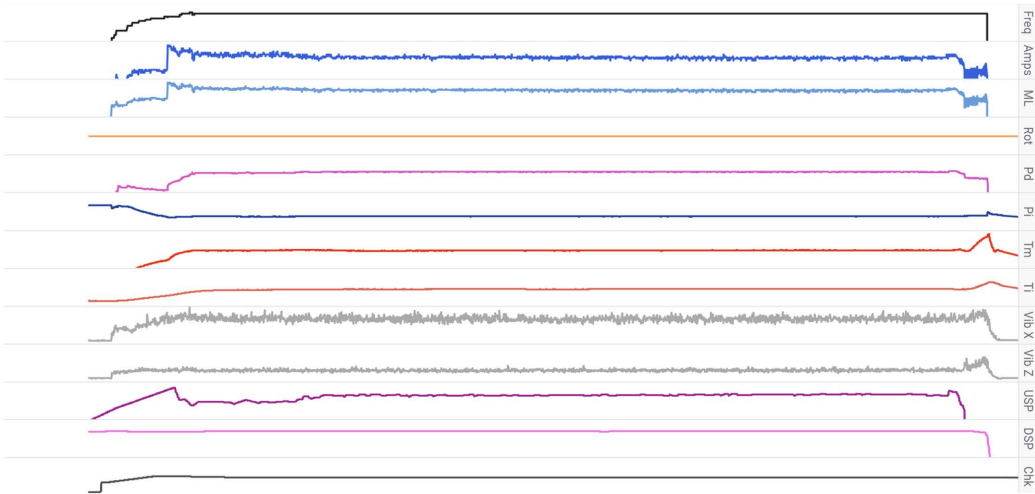


Figure 14—ESP trip due to increase in fluid specific gravity.

Diagnosis

Heavy Fluid: Increase in WC%.

Markers

Heavy Fluid: Higher Pd and lower WHP; higher motor amps and load; higher Tm and lower WHT.

Remedy

Higher frequency and bigger choke size opening.

Table 11—increase in fluid specific gravity parameters trend.

Parameter	Trend
Amps	Increase
Discharge Pressure	Increase
Motor Temperature	Increase
Wellhead Pressure	Decrease

Case #12.

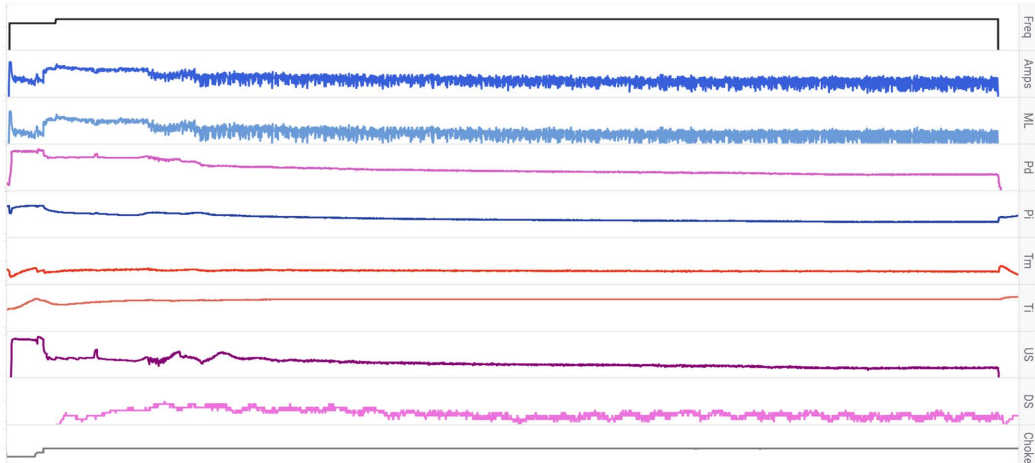


Figure 15—ESP trip due to gas lock.

Diagnosis

Gas-Lock.

Markers

Excessive Gas Production: High fluctuations in motor amps, discharge pressure, and WHP; decrease in discharge pressure.

No Flow: Low amps and intake temperature; increase in motor temperature.

Remedy

Each well - and usually trip - is different and thus requires different approach.

Table 12—gas-lock parameters trend.

Parameter	Trend
Amps	Decrease
Discharge Pressure	Decrease
Motor Temperature	Increase

Table 13—general guidelines for troubleshooting wells susceptible to gas-locking (Alsayedali, et al. 2024).

Frequency Setting	Choke Valve Opening	Advantages	Disadvantages
Low	Small	Low flowrate leading to less drawdown, higher intake pressure, and less GVF%. Relatively higher gas-handling efficiency due to the higher intake pressure.	Less heavy fluid handling capability due to the low head developed by the pump and the restriction imposed by the small choke opening. Relatively lower gas-handling capability due to operating on the left hand-side of the curve which means the ESP can trip on higher intake pressure compared to when it is operating at higher frequency.
High	Small	Relatively higher gas-handling capability which means the ESP can sustain lower intake pressure compared to when it is operating at lower frequency. The small choke valve opening can still keep the intake pressure relatively sustained.	While the higher frequency can help with the heavy fluid the small choke valve opening still leads to less heavy fluid handling capability due to the flow restriction. For wells that are susceptible to gas slugging flow behavior a smaller choke valve setting can worsen the issue.
Low	Large	Beneficial for wells with relatively high water cut and small GVF%. Sometimes the only solution for wells that experience repetitive pump-off and gas-locking events due to the lower centrifugal forces being imposed on the fluid.	While the big choke size can help with the heavy fluid the low frequency still leads to less heavy fluid handling capability due to the head developed by the pump. For some special scenarios higher head might be necessary to maintain the flow with an increase in gassy flow.
High	Large	Relatively better heavy fluid handling capability due to the high head developed by the pump and the absent of restrictions imposed by the choke valve opening. Relatively better gas-handling efficiency and improved gas-handling capability.	Very fast pressure depletion leading to lower intake pressure and higher GVF%. Higher frequency will lead to higher centrifugal forces being imposed on the fluid leading to higher phase separation.

Case #13.

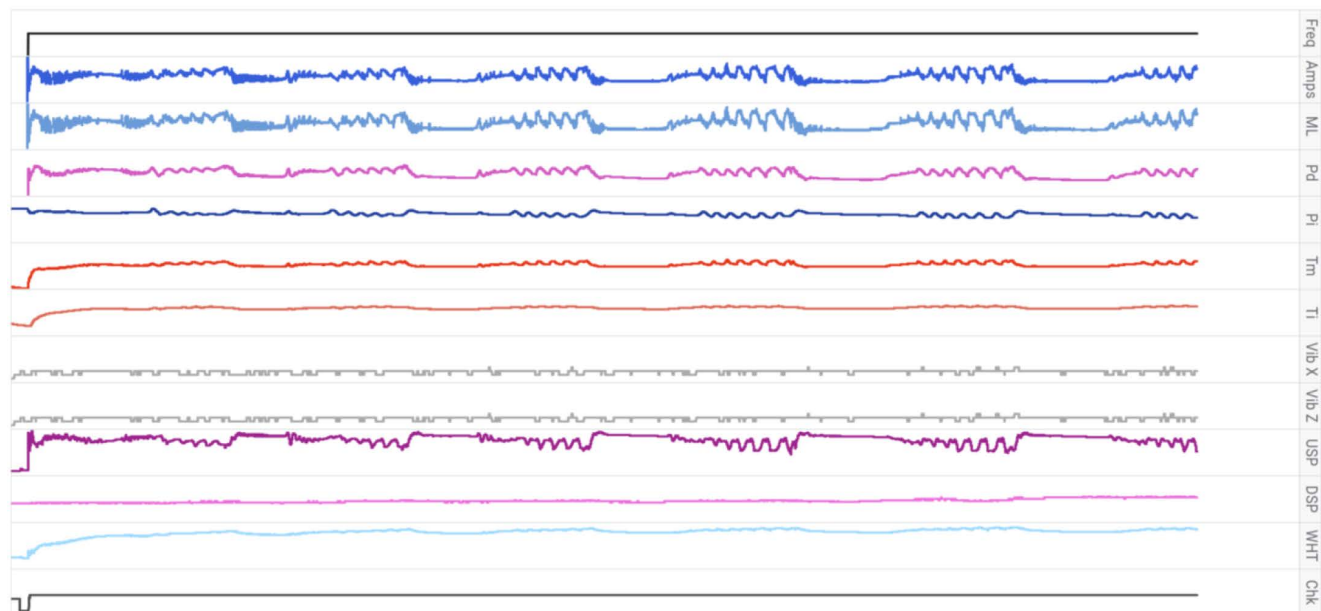


Figure 16—ESP trip due to fluctuations in fluid specific gravity.

Diagnosis

Heavy Fluid & Light Fluid: High Water Cut & High GVF%.

Markers

Heavy Fluid: High motor amps and load; high discharge pressure; parameters fluctuations indicating flow instabilities.

Gassy Fluid: Amps and other parameters fluctuations due to gas surging, sudden intake & motor temperature changes (increase due to no heat dissipation; gasses are poor heat convection fluids).

Remedy

Case-by-Case based on the percentage of WC% GVF%.

If the heavy fluid is due to workover fluid then higher frequency and higher choke opening will help with the heavy fluid and will ensure faster wellbore cleaning.

Otherwise higher frequency (for heavy fluid) and low-medium choke size opening (to control drawdown and hence GVF%) might be more suitable.

Table 14—gas-lock parameters trend.

Parameter	Trend
All Parameters	Fluctuating

ESP Troubleshooting Workflow

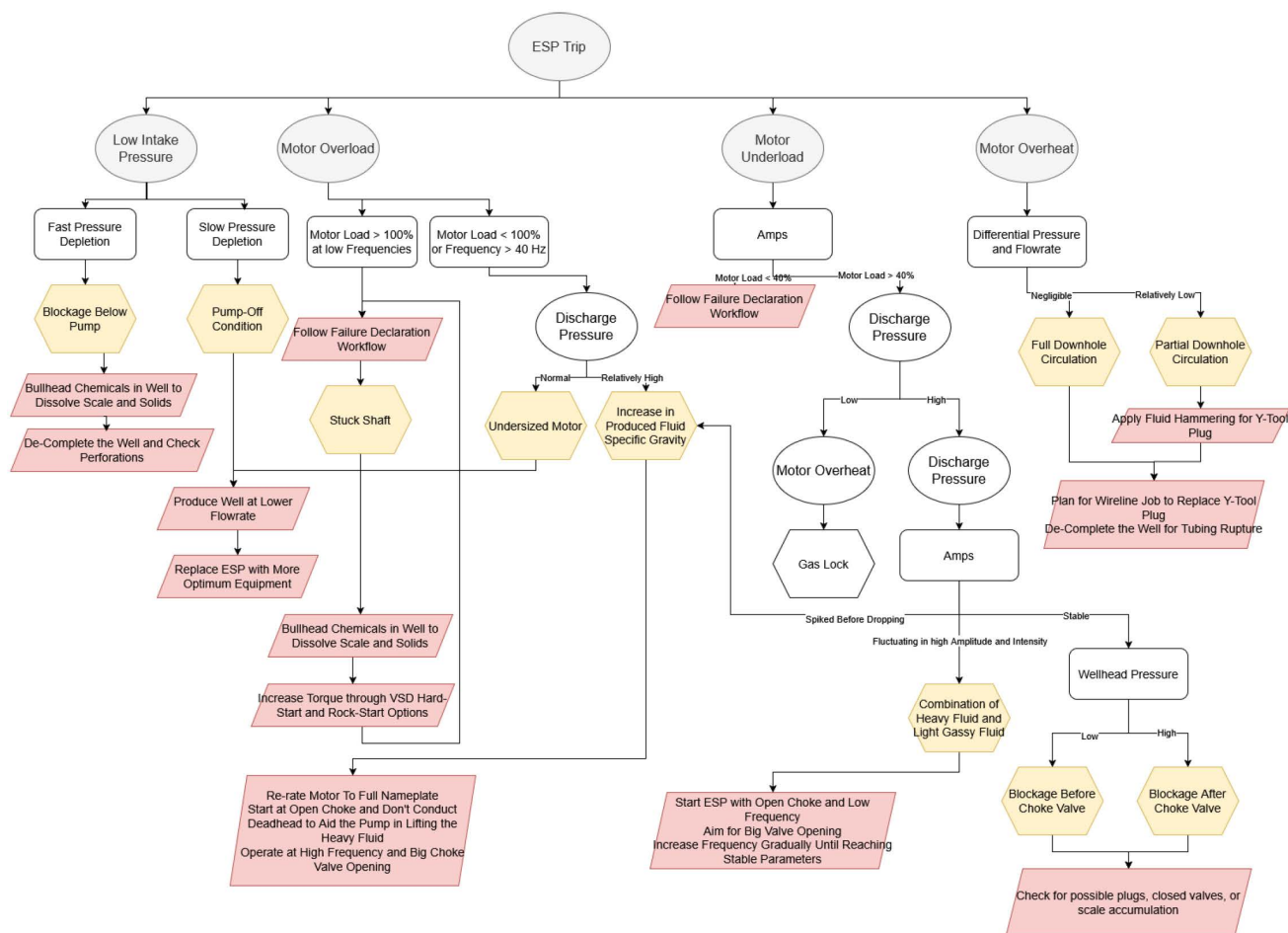


Figure 17—Recommended ESP Troubleshooting Workflow.

ESP Stuck Shaft Workflow

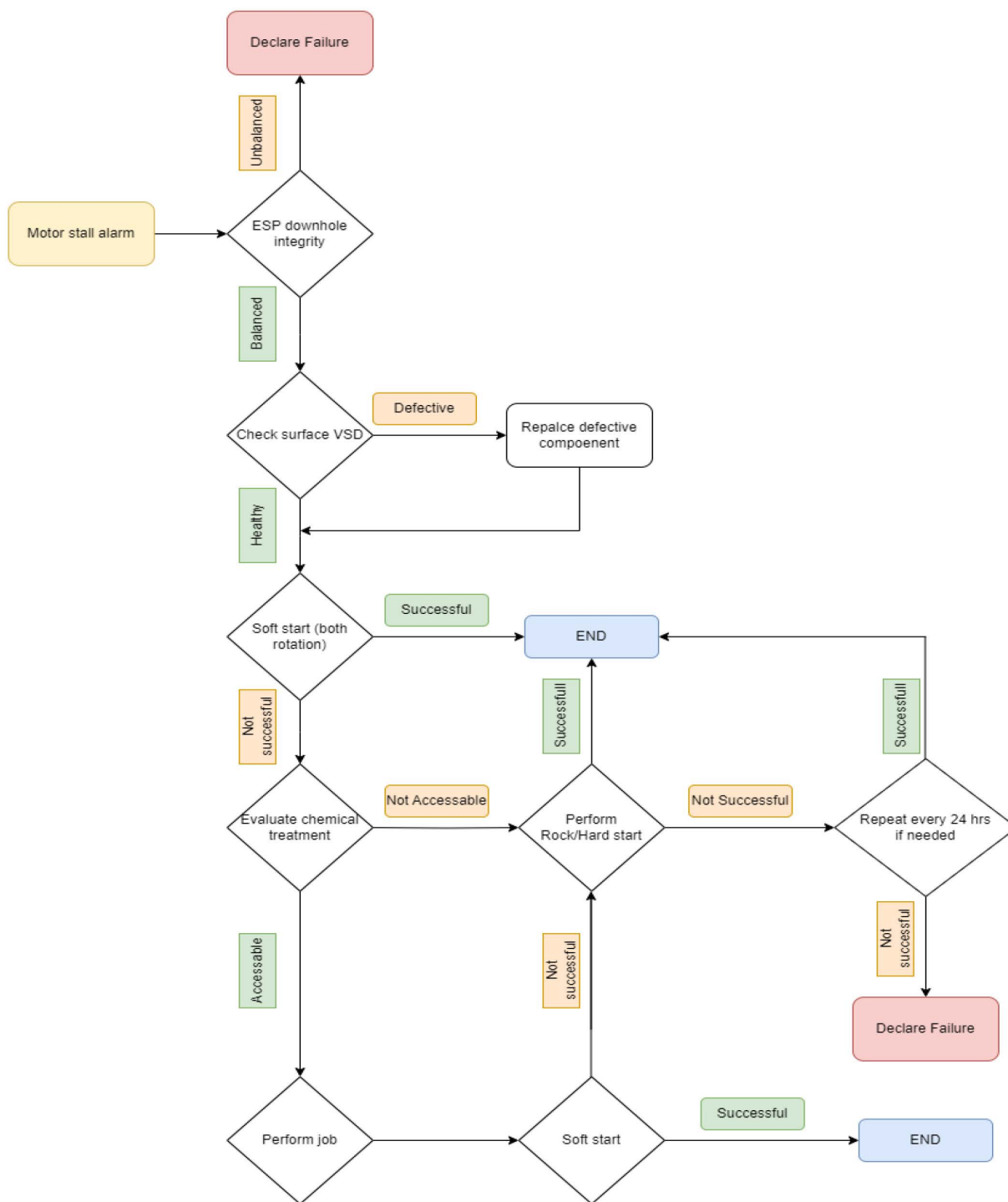


Figure 18—Recommended ESP Stuck Shaft Workflow.

ESP Failure Declaration Workflow

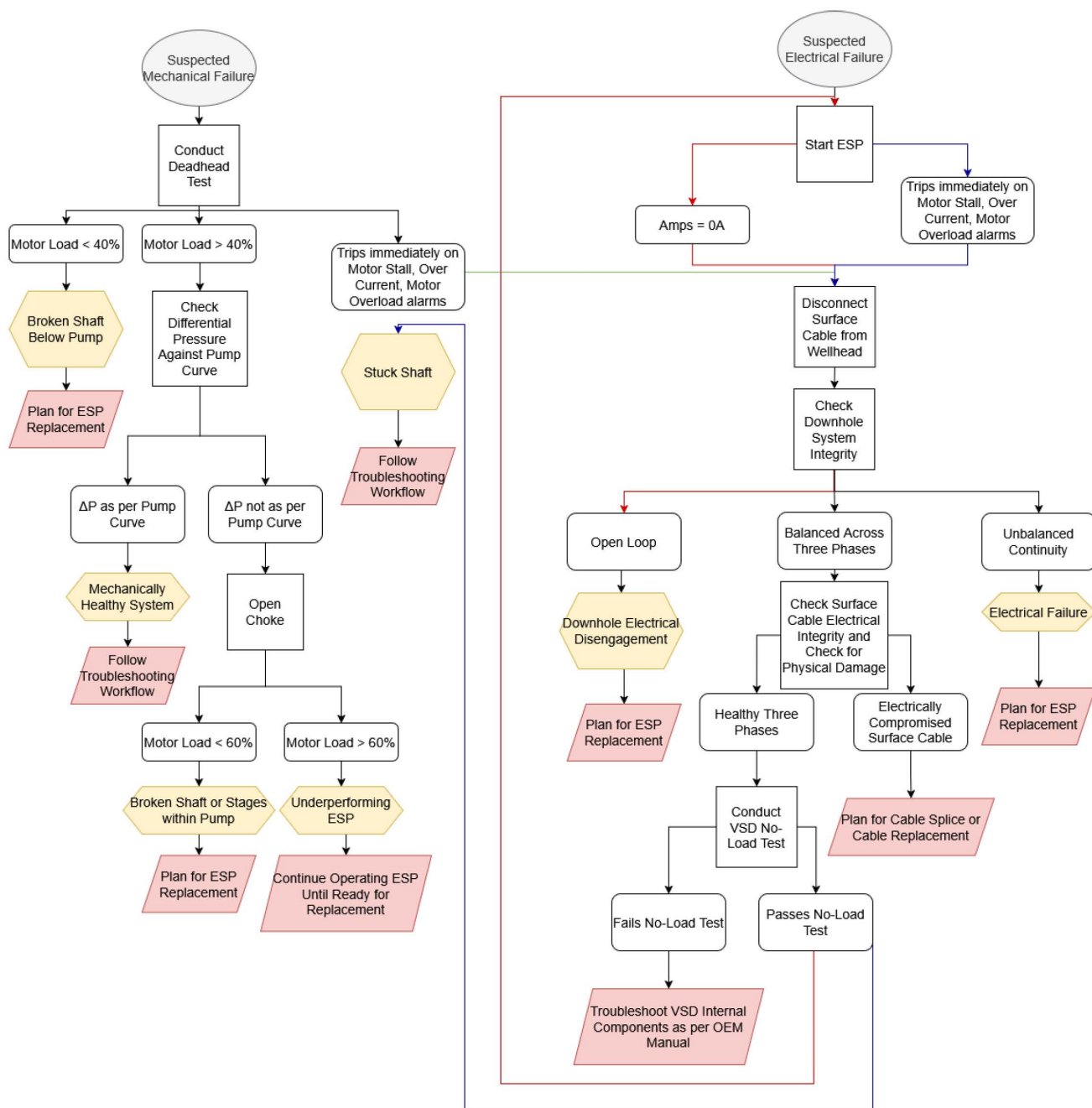


Figure 19—Recommended ESP Failure Declaration Workflow.

Recommendations

Due to the complexity of ESP systems and the different electrical and mechanical components in the system composition, they can trip due to multiple casual factors. These causal factors can vary in their complexity as well and they can be of internal cause (ESP related) and external cause (environment related). To avoid further production interruption due to the ESP trip and reduce the repetitive trips, it is recommended to follow the discussed troubleshooting workflow to ensure reaching accurate conclusion about the exact root cause of ESP trip and identify it promptly. Moreover, if all troubleshooting attempts fail to return the ESP back on-stream then it is recommended to follow the failure declaration workflow to ensure accurate ESP status classification to avoid improper classification of its status.

For future work, it is recommended to:

- Incorporate these workflows as algorithms in software tools to automate the process. The results obtained from the automation tools can be benchmarked with the manual process results to assess and validate its accuracy.
- Explore building a machine learning model and feeding it the behaviour of the subsurface and surface parameters before ESP trip for the cases discussed in this paper as a next step in the automation process.

Summary

This paper discussed the different major conditions that can result in ESP downtime. These conditions can further impact ESP mechanical and electrical integrity negatively leading to ESP failure. As such, the paper outlined two workflows that are recommended be employed by artificial lift engineers while troubleshooting ESP trips.

The methods and procedures in those workflows are pivotal in ensuring a steady and continuous decrease in deferred production and increase in ESP systems runlife by significantly reducing repetitive trips caused by improper trip identification. This significant reduction in trips ensures the longevity of ESP population as it protects their mechanical and electrical integrity. The positive impact of such procedures can be tracked through select ESP key-performance-indicators (KPI) (Al-Khalaf, et al. 2020) with immediate results in the first year and significant improvements by third year of implementation as evident by below graph.

ESP trips can be challenging to analyze and rectify; they also might require the ESP to be shut-in for extended periods for the troubleshooting duration. As such, reducing these trips through following these workflows can have a major impact on the field production strategies. It is worth noting that, ESP frequent shutdowns and start-ups (normal start/stop or tripping) can impact and impose higher stresses on the integrity of the systems (Albori, et al. 2024).

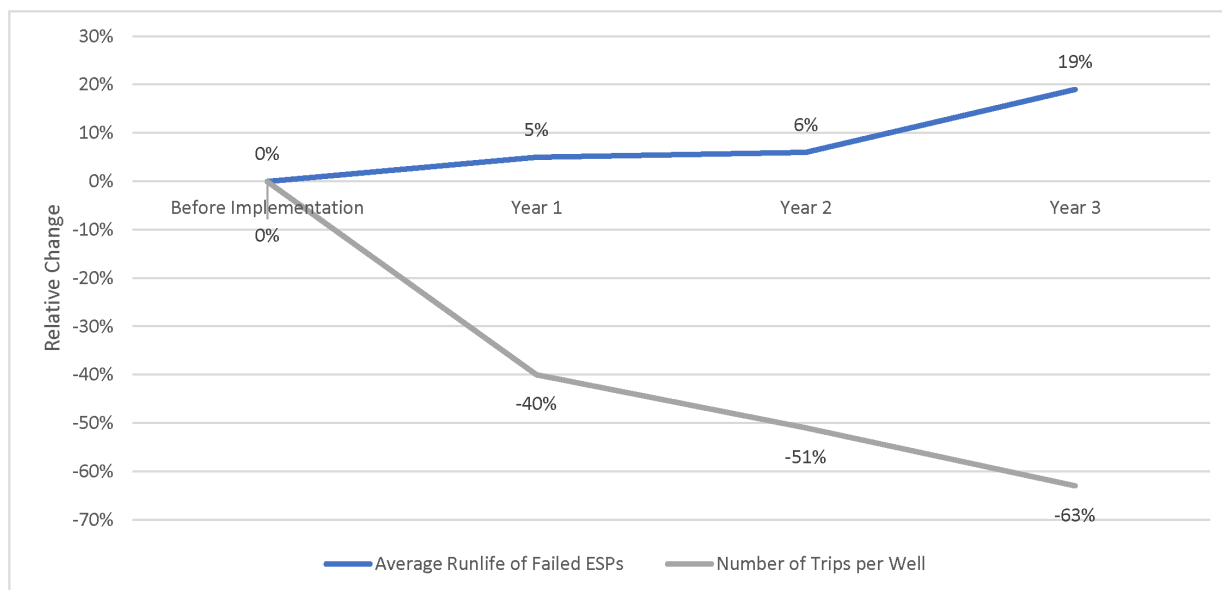


Figure 20—Expected improvements due to implementation of the discussed procedures.

Nomenclature

ESP	E lectrical S ubmersible P ump
RoR	R ecommended O perating R ange
WHP	W ellhead P ressure

VSD Variable Speed Drive
A Electrical current in "Amps"
Pi Intake pressure in "psi"
Pd Discharge pressure in "psi"
Tm Motor temperature in "°F"
Freq Operating motor frequency in "Hz"

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